

# Nicholas Ashford

Software Engineer · Distributed Systems, Cloud Infrastructure & Agentic AI

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I'm a software engineer and systems programmer with 18+ years of experience, currently CTO of Education Intelligence, the research business behind Teacher Tapp and School Surveys, where the platform is built in TypeScript, Go and Rust. I own its infrastructure end to end: microservices over message brokers, running on Kubernetes on Google Cloud. Over the past year I've also rebuilt the company to be AI-native: agents now implement our tickets, sit in our planning calls, and connect to every system in the business. I'm comfortable getting stuck into unfamiliar systems and tracking problems down to their root cause.

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## Highlights

- **Agentic delivery:** built a multi-agent software factory pipeline, with productivity gains that have let the business absorb natural attrition without replacement hires.
- **Systems programming:** fault-tolerant, idempotent microservices in Go and Rust that absorb extremely spiky traffic.
- **Infrastructure:** owned cloud infrastructure across roles in AWS, Azure and Google Cloud, built with Terraform and Kubernetes.
- **Custom integrations:** energy-provider integrations at Switcher.ie and uSwitch, on-prem data extraction inside schools, MCP proxies for closed SaaS products.
- **Voice agent:** joins project management calls and raises contextualised tickets from the conversation.
- **MCP infrastructure:** AI in the hands of everyone in the business, with custom MCP connectors and proxies for the systems that lacked them.
- **Early CI:** built an automated pull-request-to-production pipeline at uSwitch, before mainstream CI tooling existed.

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## Experience

### Lead Engineer & CTO

#### Education Intelligence (Teacher Tapp & School Surveys)

January 2021 – Present

*Education research business running the UK's largest daily teacher survey panel, with thousands of teachers responding every day.*

- Rebuilt the Teacher Tapp app from an MVP into TypeScript and React Native on a distributed backend.
- Took the School Surveys MVP to a production-grade platform in TypeScript and Rust.
- Architected both platforms as idempotent microservices communicating over message brokers, with queue consumers written in Go.
- Migrated the Teacher Tapp infrastructure from Azure to Google Cloud.
- Now responsible for two distributed, fault-tolerant platforms and nine TypeScript applications across the two businesses.
- Designed and built a multi-agent pipeline: a workflow of agents implements tickets from specifications and acceptance criteria, raises pull requests, and ships only after automated tests and human review.
- Built a voice agent that joins project management calls, listens to the discussion and raises contextualised tickets, replacing the Kanban board.
- Rolled AI out to every member of the business and connected its systems to Claude via MCP, including a proxy service that exposes MCP connectors for SaaS products that lack them.
- Built and led a team of engineers, mentoring each of them through to more senior roles.
- Responsible for security: protecting sensitive survey and school data, hardening infrastructure, and remediating findings from security testing.
- Established the end-to-end product process: specification, design, build, release and analytics.

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### Senior Software Engineer

#### Switcher.ie

February 2019 – January 2021 · Dublin

- Part of a small team of senior engineers building the service-oriented platform behind Ireland's leading comparison site, handling lead capture from the website.
- Designed and built the framework for distributing leads to the utility providers.
- Owned the infrastructure: preview, staging and production environments and their monitoring, built with Terraform and Kubernetes on AWS.

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## Lead Software Engineer

### Assembly Edu

January 2016 – March 2019 · London

- Helped establish the company: structured as a 50/50 partnership between NEON's investment arm and an education provider, with the platform built from the ground up.
- Designed and built the MIS Connector, a .NET ETL system running on servers inside schools, extracting sensitive management information data and syncing it to the cloud.
- Wrote the cloud processing pipeline that turned raw school data into a queryable database, consumed by third-party education providers for analysis.
- Owned the infrastructure (AWS) and a substantial share of the Ruby platform.

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## Senior Developer

### NEON Adventures

January 2014 – January 2017 · London & New York

- Consulting and tech leadership on client projects for a venture studio, spanning charities and consumer brands.
- Built interactive 3D visualisations for a professional cycling team's website, at a time when 3D in the browser was still rare.

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## Senior Graphics Programmer

### Sony Computer Entertainment Europe

February 2012 – January 2014 · London

- Graphics programmer on Harry Potter titles for PlayStation 3 (C++/GPU), on the team responsible for special effects and integrating the art-tool pipeline.
- Built a real-time fluid simulation for the potion cauldron, implementing the Navier-Stokes equations in shaders on the PS3 GPU.
- Implemented augmented-reality features using the PlayStation Eye, projecting a handheld device into in-game 3D environments.

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## Senior Developer

### Forward Internet Group (uSwitch)

July 2008 – February 2012 · London

- Full-stack development on uSwitch.com's gas and electricity comparison products: lead capture, processing and integrations with energy providers.
- Built the company's internal deployment system: pull requests from up to 100 engineers triggered an automated pipeline through to production.
- Founded and ran an iOS games studio within Forward's incubator: hired the team, then led development, release and marketing of a C++ iOS game over the course of a year.

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## Skills & Technologies

- **Cloud & infrastructure:** AWS, Google Cloud Platform, Azure, Terraform, Kubernetes, event-driven architecture, message brokers, CI/CD pipelines, preview/staging/production environment provisioning, monitoring
- **AI & agentic engineering:** LLM orchestration and prompt engineering, agentic development workflows, multi-agent pipelines, MCP connector and proxy development, generative AI, machine learning, PyTorch
- **Languages:** TypeScript, Go, Rust, C++, Python, JavaScript, C#, Ruby, Java, SQL
- **Graphics & simulation:** GPU and shader programming, real-time fluid simulation, 3D mathematics, augmented reality
- **Data & integrations:** ETL pipelines, on-premise data extraction, third-party API integrations, Microsoft SQL Server, large-scale data processing
- **Web & mobile:** React, Node.js, .NET, iOS, Android
- **Leadership & practice:** team building and mentoring, agile delivery, test-driven development, service-oriented architecture

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## Location

British citizen currently based in Dubai, with EU work rights through spouse.